Mode veering in weakly coupled systems

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Abstract
The phenomenon of mode veering occurs in systems with a varying parameter. As the parameter varies, so do the natural frequencies. When two natural frequencies approach each other they often veer apart, instead of crossing. The veering is accompanied by rapid variations in the eigenvectors. This phenomenon is analysed in this paper with the focus being on weakly coupled systems of modes or oscillators. The system has any number of degrees of freedom or modes. The system is defined to be uncoupled when the motion in all but one of the modes is blocked. A small stiffness, mass or gyroscopic parameter is assumed to couple the uncoupled-blocked modes. The natural frequencies of the uncoupled system depend on the variable parameter and can cross at certain critical frequencies. The natural frequencies of the coupled system are seen to veer at these critical frequencies in the presence of arbitrarily small coupling, with the eigenvectors rotating and swapping from one branch to another. The separation of the branches around the critical frequencies is seen to depend on the coupling parameter. Examples including a 2 degree of freedom system, a multi-degree of freedom system and a plate with an attached variable oscillator are presented to illustrate the results.

1 Introduction

Mode veering typically refers to the phenomenon associated with the eigenvalue loci for a system with a variable parameter: two branches approach each other and then veer away and diverge instead of crossing. The veering is accompanied by coupling between modes and rapid variations in the eigenvectors. Some of the first evidence of coupling between modes was provided by Webster [1], Petyt and Fleisxer [2] and Nair and Durvasula [3]. In [3] frequency loci for varying geometric parameters of plates of different geometry, stiffness and boundary conditions were discussed. It was shown that for symmetric configurations the frequency curves diverge from each other. The authors referred to this behaviour as "quasidegeneracies". Later Leissa [4], referring to the eigenfunctions, described the phenomenon using these words: "*... must undergo violent change – figuratively speaking, a dragonfly one instant, a butterfly the next, and something indescribable in between. Thus the results appear strange from an aesthetic point of view*". He then stated that "*In numerous places in the literature of eigenvalue problems of mathematical physics one finds curves which approach each other and suddenly veer away. The author postulates that this ugly behavior may be the result of approximation in the representation of physical reality*". Since then, the phenomena has been commonly referred to as veering since, under certain coupling conditions, two or more curves describing the eigenvalue loci against a parameter of the system veer away and diverge instead of crossing - a schematic representation is given in Figure 1. Although Leissa showed that, for continuous models, curve veering can be an "aberration" introduced when natural frequencies and mode shapes are obtained using approximate methods associated with discretised models, it has been

verified later using exact solutions [5] that curve veering in continuous models is a “representation of physical reality”. Evidence of veering phenomena was also shown, for example, in pressure vessel [6] and cable dynamics [7,8]. Recently, an experimental demonstration of veering of a stressed framed structure was given in [9], where experimental results were successfully compared with those predicted by a Finite Element model. One of the earlier veering criterion was formulated by Perkins and Mote [5]. In [5] a perturbation technique was applied in a straightforward way to analyse the dependence of the eigenvalue loci on a system parameter, both in the case of continuous and discrete systems. In particular in [5] it was demonstrated that veering could not be an aberration. Later, Pierre [10] used a modified perturbation approach to overcome failings of Perkins and Mote’s technique in the vicinity of the point of maximum coupling. A number of modified perturbation approaches have been proposed, and veering phenomena in dynamics have been treated in many papers. Veering can also be investigated considering the eigenvectors. To avoid crossing, the eigenvectors rapidly rotate in their plane. This aspect was studied in [11] for a discrete three degrees of freedom system. An analysis of veering using a geometric approach was given by Triantafyllou [12]. A geometric formulation of the problem was also given by Arnold [13] in terms of ellipsoids associated to positive definite symmetric eigenvalue problems, and later by Vidoli and Vestroni [14].

Coupling between modes and veering occurs in simple structural waveguides such as rods, plates and cylinders. This is particularly true at higher frequencies, when the wavelengths of interest become small compared to the length of the structure. As examples, coupling between bending and torsional motion of uniform beams was studied by Timoshenko [18], and later by Bishop and Pierce [19].

Continuous structures with added sub-systems can exhibit mode veering. Examples abound in the literature, in particular concerning simple structural components such as beams and plates carrying spring-mass systems, e.g. [15, 16]. Recently, in [17] the free vibrations of a spring-mass system attached to a cantilever plate were investigated in terms of the natural frequency loci of the coupled system.

The phenomenon has also been observed in the vibrations of disordered structures. The presence of small irregularities in weakly coupled periodic systems affects the system behaviour in terms of vibration mode localisation (vibration modes are “confined” to a limited region of the system) associated with veering of the frequency loci. The phenomenon was predicted in an early paper by Anderson [20] studying random lattices. In the context of vibration of structures, two early studies, where mode localisation was discussed, were given by Hodges [21], and Hodges and Woodhouse [22]. In the latter, weak and strong localisation, according to the magnitude of internal coupling, was shown. The relation between veering and mode localisation in periodic structures perturbed by a disordered parameter was investigated by Pierre in [23]. In this paper, Pierre claimed that veering does not occur when coupling is strong. This was confirmed by Pierre in [24], where the effects of disorder in chains of coupled oscillators were investigated using a probabilistic approach. The result was later criticised by Chen and Ginsberg [25]. Veering of frequency loci associated with mode localisation was also studied and quantified by Lee and Mote [26] for a translating string coupled to a spring-mass-damper constraint, where frequency loci and mode localisation were evaluated as functions of the constraint location. It is worth pointing out that the behaviour in terms of localisation at high frequency and for higher modes could be different from that at low frequency and for low modes [27].

Although the phenomenon has been investigated by many authors, few papers have tackled the problem in a general way and most of the literature concerning mode veering deals with specific systems. Moreover, except from some simple discrete cases, the complicated effects of veering and coupling between modes cannot be modelled theoretically, and a complete analysis typically requires a formulation on a case-by-case basis. Following previous work by current authors concerning wave propagation phenomena [28], the purpose of this work is to investigate and classify veering and the circumstances under which it occurs in general terms in linear and lossless structural systems.

In section 2 mode coupling is introduced and illustrated in general terms. The natural frequencies of the uncoupled system depend on a variable parameter and can cross at certain critical frequencies. The natural frequencies of the coupled system are seen to veer close to these critical frequencies in the presence of arbitrarily small coupling, with the eigenvectors rotating and swapping from one branch to another. A coupling parameter is defined and the separation of the branches around the critical frequencies is seen to

depend on this parameter. The main characteristics of veering are discussed in order to deduce general features of coupling between modes. Some simple numerical examples are considered in order to illustrate the behaviour. Then a more complicated example of a plate with an attached mass-spring oscillator is presented. Finally some concluding remarks are given in section 4.

2 Mode veering: weakly-coupled random oscillators

The equations of motion of free vibration for an undamped system can be written in discrete form as

$$\left[\mathbf{K}(\mathbf{p}) - \lambda \mathbf{M}(\mathbf{p})\right]\mathbf{x} = 0 \tag{1}$$

where **K** and **M** are the mass and stiffness matrices, **p** is a vector of random (variable) parameters, the eigenvalues λ are the squares of the natural frequencies and the vector **x** is an eigenvector of degrees of freedom (DOFs), i.e. the mode shape. As the parameters vary, the eigenvalues follow loci in the $(\mathbf{p}, \lambda)$ space. Veering occurs when two eigenvalues approach closely as a parameter increases and then veer apart, with consequent rapid changes in the eigenvectors, as illustrated in Figure 1. Note that the eigenvectors associated with each branch of the spectrum are in essence exchanged. In general, the behavior is very complicated and it is difficult to draw any broad conclusions. In this section various simplifying assumptions are made which allow analytical solutions to be developed in the veering region. The analysis follows similar lines for other situations, and is described elsewhere.

2.1 Weak coupling and the uncoupled-blocked system

There is assumed to be just one random stiffness parameter *p*, which affects one or more elements on the diagonal of the stiffness matrix. (The analysis for the case of a random parameter in the mass matrix follows similar lines.) The individual DOFs might represent oscillators or modes of the uncoupled system, the off-diagonal terms representing coupling between these modes or oscillators. ***Weak coupling*** is

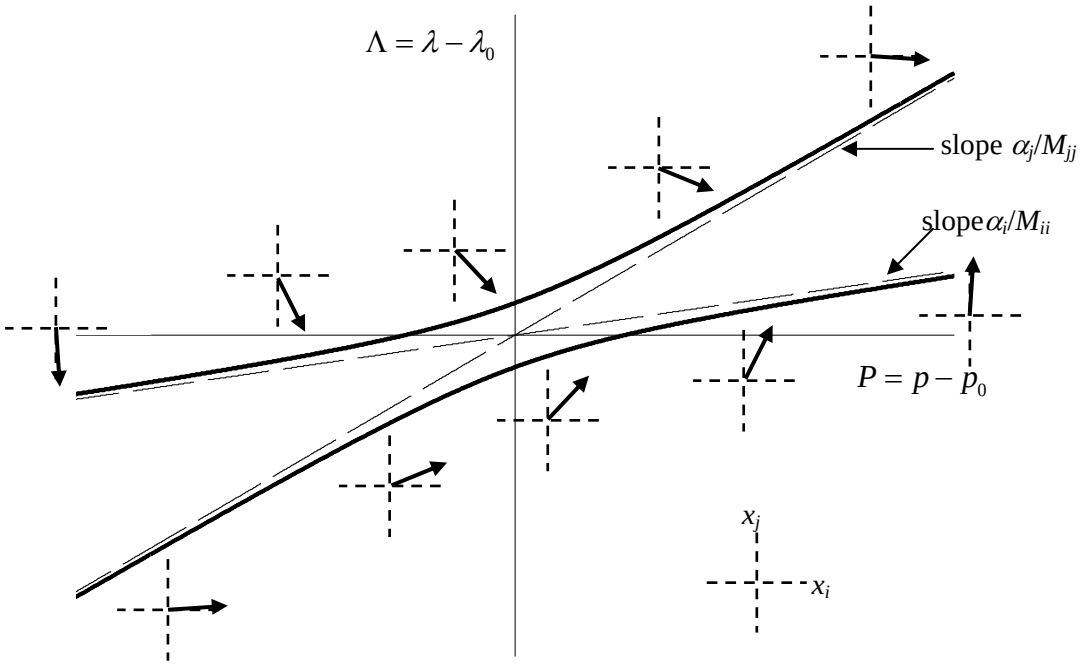


Figure 1: Veering: dispersion curves for _ _ _ uncoupled-blocked and _____ coupled DOFs x_i and x_j. Inlays show directions of eigenvectors in (x_i, x_j) plane.

assumed, and is defined such that the off-diagonal terms are of order ε compared to the elements of the diagonal terms, which are of order $O\left(K_{jj}\right) \sim O\left(\lambda M_{jj}\right) \sim O(1)$. The ***uncoupled*** system is defined by ***blocking*** all DOFs apart from one: the ***uncoupled-blocked*** eigenvalues are therefore $\lambda_j^0 = K_{jj}(p)/M_{jj}$. In summary, then, the off diagonal elements of the matrix $\mathbf{K}(p) - \lambda\mathbf{M}$ are $O(\varepsilon)$, while the diagonal elements K_{jj} and λM_{jj} are $O(1)$. Note, however, that their difference $\left(K_{jj} - \lambda M_{jj}\right)$ can be $O(\varepsilon)$ or smaller for values of λ around the blocked eigenvalues λ_j^0 – it is this that causes veering. The eigenvalues are such that $\det\left(\mathbf{K}(p) - \lambda\mathbf{M}\right) = 0$.

2.2 A 2-DOF system

Consider first, for simplicity, the 2-DOF system shown in Figure 2. The variable parameter is the stiffness *p*, the coupling element is a weak spring ε. The eigenvalues are the solution to

$$\begin{vmatrix} k_1 + \varepsilon - \lambda m_1 & -\varepsilon \\ -\varepsilon & p + \varepsilon - \lambda m_2 \end{vmatrix} = 0 \tag{2}$$

while the uncoupled-blocked eigenvalues are

$$\lambda_1^0 = \left(k_1 + \varepsilon\right)/m_1; \quad \lambda_2^0 = \left(p + \varepsilon\right)/m_2 \tag{3}$$

The uncoupled-blocked eigenvalues are shown as functions of *p* in Figure 3. These loci – the skeleton of the uncoupled-blocked system – intersect at the critical point

$$p_0 = \frac{k_1 m_2 + \varepsilon\left(m_2 - m_1\right)}{m_1} \tag{4}$$

at which $\lambda_1^0 = \lambda_2^0 = \lambda_0$. Now the substitution

$$P = p - p_0; \quad \Lambda = \lambda - \lambda_0 \tag{5}$$

is made, so that the critical point is at $P = \Lambda = 0$. The eigenvalues are now the solutions to

$$\begin{vmatrix} -\Lambda m_1 & -\varepsilon \\ -\varepsilon & P - \Lambda m_2 \end{vmatrix} = 0 \tag{6}$$

and are given by

$$\Lambda = \frac{P \pm \sqrt{P^2 + 4\frac{m_2}{m_1}\varepsilon^2}}{2m_2} \tag{7}$$

These are also shown in Figure 3. For values of *p* far from p_0 (so that $P^2 \gg 4m_2\varepsilon^2/m_1$) the eigenvalues

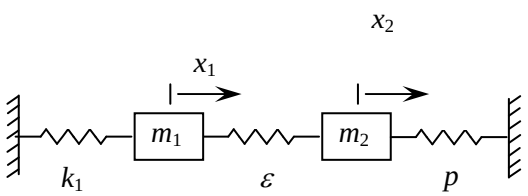


Figure 2: 2-DOF weakly coupled system.

are just perturbations of $O(\varepsilon)$ from the uncoupled-blocked values: the higher eigenvalue increases slightly, the lower eigenvalue decreases slightly. However, around the critical point the diagonal terms are $\leq O(\varepsilon)$, so that the coupling terms, no matter how weak, are the same order of magnitude or larger than the diagonal terms. Veering occurs: the two eigenvalues can never cross because of the term involving ε^2 in the surd. At the critical point (P = 0) the eigenvalues are

$$\Lambda = \pm \frac{2}{\sqrt{m_1 m_2}} \varepsilon \tag{8}$$

There is thus a region of width $O(\varepsilon)$ around the critical point in the (P,Λ) plane. The insets in Figure 3 also indicate the directions of the eigenvectors. Far from the critical point the eigenvectors, defined by the amplitudes of motion of the two masses (x_1,x_2), are perturbations of $O(\varepsilon^2)$ from the uncoupled-blocked values of (±1,0) and (0,±1). However, they change rapidly around the critical point.

2.3 Weakly coupled MDOF systems

For the case of a weakly-coupled MDOF system with a single, variable stiffness parameter p, as described in section 2.1, the eigenvalues are found from the solutions to

$$\left|\mathbf{K}(p) - \lambda \mathbf{M}\right| = 0 \tag{9}$$

The uncoupled-blocked eigenvalues $\lambda_j^0 = K_{jj}(p)/M_{jj}$ are found by setting the diagonal terms to zero. Far from any critical point where two uncoupled-blocked eigenvalues intersect (the case of three intersecting eigenvalues is not considered here) the characteristic equation is

$$\prod_{jj=1}^{n}\left(K_{jj}(p) - \lambda M_{jj}\right) + O(\varepsilon) = \prod_{jj=1}^{n}\left(\lambda_j^0(p) - \lambda\right) M_{jj} + O(\varepsilon) = 0 \tag{10}$$

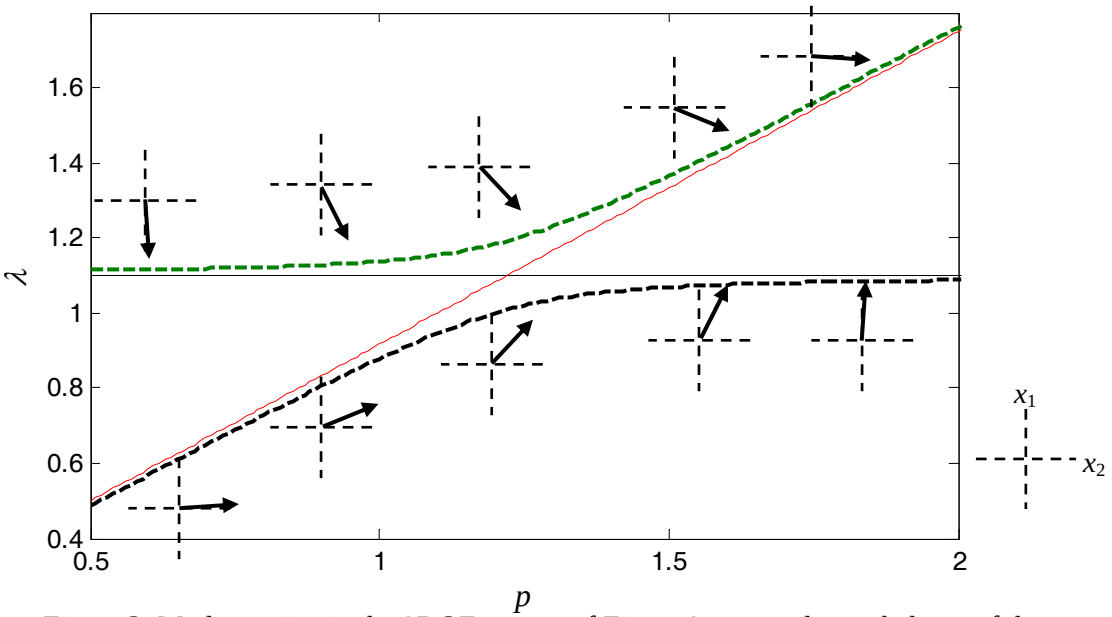


Figure 3: Mode veering in the 2DOF system of Figure 2: ___ and ___ skeleton of the uncoupled-blocked system; - - - and - - - eigenvalues of the coupled system. Inlays show directions of eigenvectors.

The eigenvalues are then just perturbations of $O(\varepsilon)$ from the uncoupled-blocked values. Now suppose that for some parameter value p_0, two uncoupled-blocked eigenvalues (for DOFs *i* and *j*, say) cross. The eigenvalues associated with the loci that cross are (to $O(\varepsilon^2)$) the solutions to

$$\begin{vmatrix} K_{ii}(p)-\lambda M_{ii} & -\varepsilon \\ -\varepsilon & K_{jj}(p)-\lambda M_{jj} \end{vmatrix} = 0 \tag{11}$$

while the remaining eigenvalues are still perturbations of $O(\varepsilon)$ from the uncoupled-blocked values. Taking a first order Taylor series expansion of $K(p)$ about p_0 and making the substitutions of Equation (5) leads to

$$\begin{vmatrix} \alpha_i P-\Lambda M_{ii} & -\varepsilon \\ -\varepsilon & \alpha_j P-\Lambda M_{jj} \end{vmatrix} = 0 \tag{12}$$

where $\alpha = \partial K(p_0)/\partial p$. The eigenvalues of the coupled system are therefore

$$\Lambda = \frac{\left(\alpha_j M_{ii}+\alpha_i M_{jj}\right)P \pm \sqrt{\left(\alpha_j M_{ii}-\alpha_i M_{jj}\right)^2 P^2 + 4M_{ii}M_{jj}\varepsilon^2}}{M_{ii}M_{jj}} \tag{13}$$

In summary, if the uncoupled-blocked eigenvalues cross at $P = 0$ then the eigenvalues of the coupled system veer: they do *not* cross, equalling $\Lambda = \pm 2\varepsilon/\sqrt{M_{ii}M_{jj}}$ when $P = 0$. In addition, $\Lambda = 0$ when

$$P = \pm\sqrt{\frac{M_{ii}M_{jj}}{\alpha_i M_{jj}+\alpha_j M_{ii}}}\varepsilon \tag{14}$$

Thus even for arbitrarily weak coupling, veering occurs in a region of $O(\varepsilon)$ around the critical points where the uncoupled blocked eigenvalues cross. This is illustrated in Figure 1. Note also the directions of the eigenvectors of Equation (12), and how they rapidly change direction in the (x_i,x_j) plane – the motion of the mode associated with each branch changes rapidly from predominantly involving one DOF to predominantly involving the other.

2.3.1 Numerical example

In this section the case of a MDOF spring-mass chain is considered. There are *n* masses, the *i*th mass m_i is connected to its neighbours by springs of stiffness k_i and k_{i+1} and to ground by an additional spring k_0 as shown in Figure 4. In the numerical example the spring k_n is small (i.e. a weak coupling between m_{n-1} and m_n) and $k_{n+1} = p$ is the variable parameter. This system thus consists of a $(n - 1)$ DOF system with fixed parameters weakly coupled (through the spring k_n) to an oscillator m_n with a variable stiffness *p*.

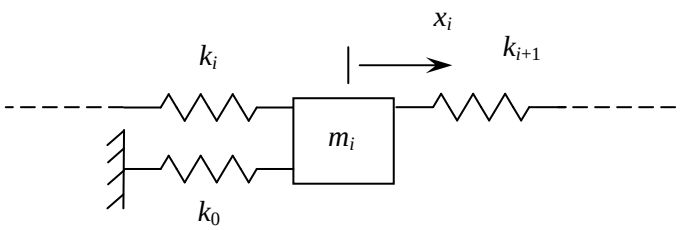


Figure 4: The *i*th mass in a mass-spring chain of *n* masses.

The uncoupled-blocked eigenvalues for the system with fixed parameters are found by an eigenanalysis of the system comprising masses 1 to $n-1$ and springs 1 to n – these do not depend on p. For mass m_n the uncoupled-blocked eigenvalue is

$$\lambda_n^{(0)} = \frac{k_n + k_{n+1}}{m_n} = \frac{k_n + p}{m_n} \tag{15}$$

These are shown in Figure 5(a). The critical points are the intersections of these uncoupled-blocked eigenvalues. The eigenvalues of the coupled system veer around these critical points as shown in Figure 5(b). The eigenvector of the lowest mode for small p (< 0.5) and the highest mode for large p (> 4.5) are essentially identical, involving motion predominantly of mass m_n.

3 Continuous system: plate coupled to a variable oscillator

In this section a more complicated example of coupling between eigenmodes is illustrated. The system comprises a simply supported plate to which a 1 DOF oscillator is attached at point (x_0,y_0) through a weak coupling spring of stiffness ε as shown in Figure 6. The stiffness p of the oscillator is variable. The amplitudes y_j of the modes of vibration of the plate alone are taken as the DOFs of the plate. The jth mass-normalized mode shape is

$$\phi_j(x,y) = \frac{2}{\sqrt{\rho h L_x L_y}} \sin\frac{m_j \pi x}{L_x} \sin\frac{n_j \pi x}{L_y} \tag{16}$$

where m and n are integers, ρ is the material density and L_x and L_y are the lengths of the sides of the plate. The uncoupled-blocked eigenvalues of the plate are

$$\lambda_{p,j}^{(0)} = \lambda_{p,j} + \varepsilon\phi_j^2(x_0, y_0) \tag{17}$$

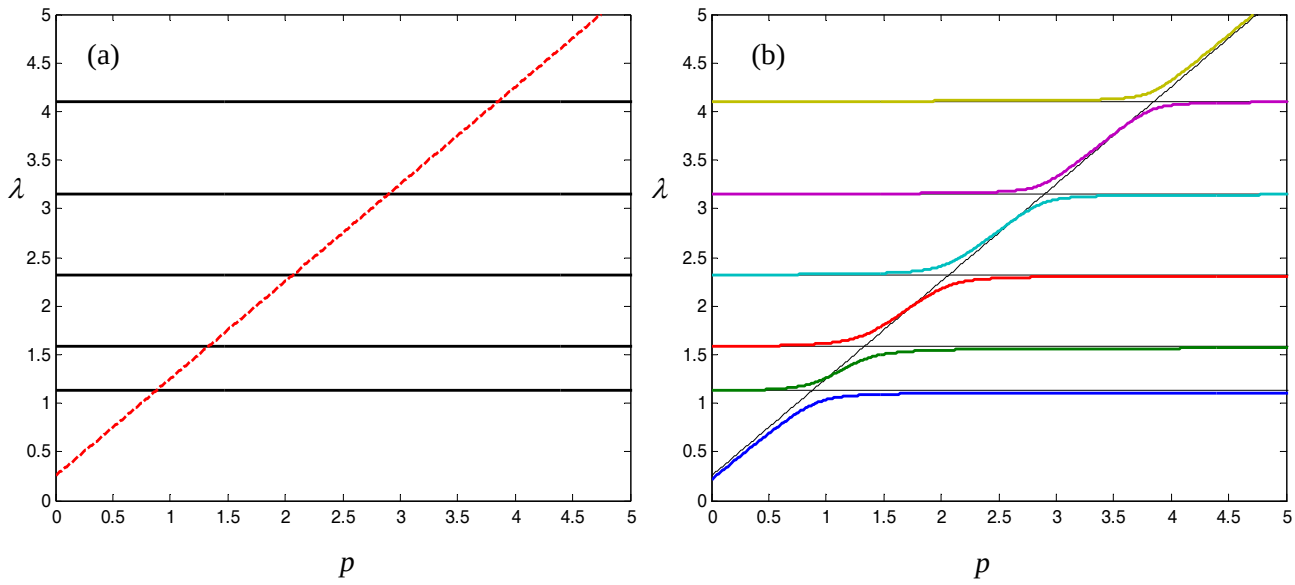


Figure 5: Veering in a 6-DOF system: $m_i = 1$; $k_0 = 1$; $k_i = k_0(1+i/6)/2$; $k_n = k_0/4$; k_{n+1} is a variable stiffness. (a) Skeleton: eigenvalues of the uncoupled-blocked system, ____ eigenvalues of fixed system; - - - - eigenvalue of system comprising m_n, k_n and k_{n+1}. Note intersections at the critical values of p. (b) Eigenvalues of coupled system. Note veering around the critical points.

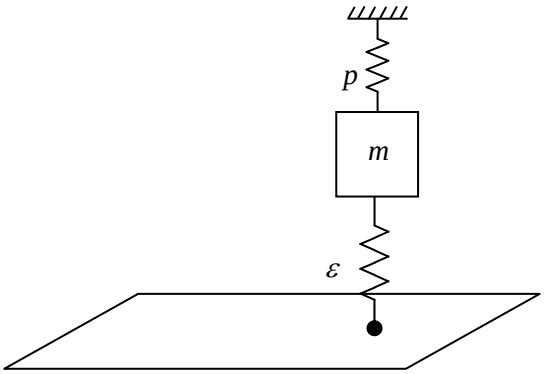


Figure 6: Simply supported plate with an attached oscillator.

where $\lambda_{p,j}$ is the *j*th eigenvalue of the plate alone. The uncoupled-blocked eigenvalue of the oscillator is

$$\lambda_o^{(0)} = \frac{p+\varepsilon}{m} \tag{18}$$

When coupled, the mass matrix is diagonal but there are small off-diagonal terms $-\varepsilon\phi_i(x_0,y_0)\phi_j(x_0,y_0)$ which couple blocked modes *i* and *j* in the plate.

Figure 7 shows the loci of the natural frequencies of the uncoupled-blocked and coupled systems as functions of *p*. The non-dimensional parameters

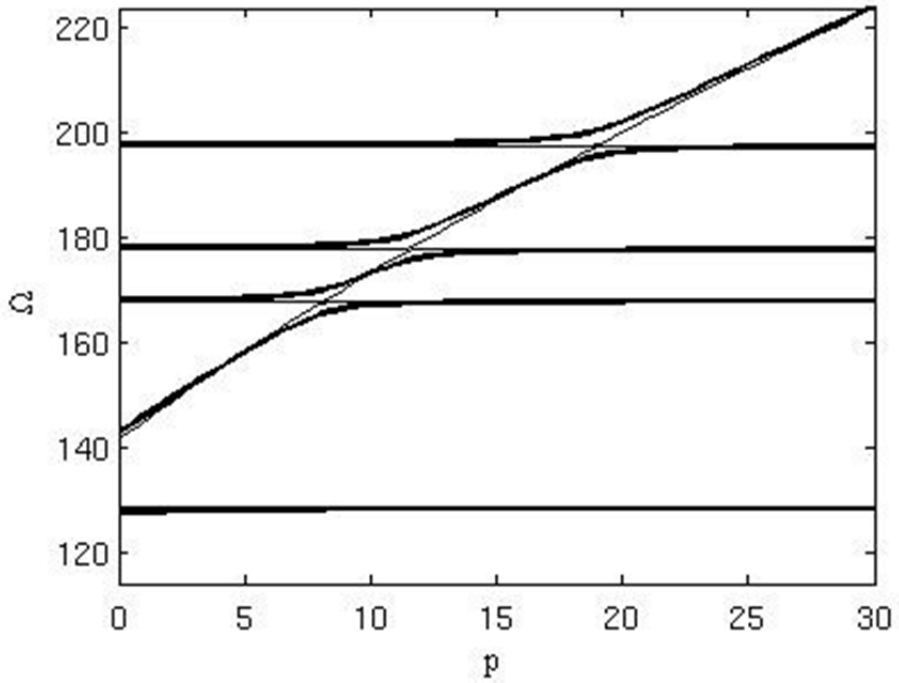


Figure 7: Veering in a simply supported plate with an attached oscillator:
$L_y/L_x = 1.5;\ m_a = 0.008;\ \varepsilon_a = 20;\ (\bar{x},\bar{y}) = (0.43, 0.22).$

$$\Omega = \omega L_x^{\,2}\sqrt{\frac{\rho h}{D}};\quad \varepsilon_a = \frac{\varepsilon L_x^{\,2}}{D};\quad m_a = \frac{m}{\rho h L_x L_y};\quad \bar{x} = \frac{x_0}{L_x};\quad \bar{y} = \frac{y_0}{L_y}$$

are introduced. Again it can be noticed that far from the critical points the natural frequencies are almost those of the uncoupled system, while close to the critical points coupling occurs and the eigenvalues veer. Furthermore, if the oscillator is attached on a nodal line, the corresponding eigenmodes do not couple and eigenvalues do not veer.

4 Discussion

The analysis above concerned weak stiffness coupling. The cases of weak mass or gyroscopic coupling, or any combination of the three, follow the same lines: the weak coupling results in a perturbation of the uncoupled-blocked eigenstructure and rapid veering around the critical points.

The case of strong coupling – when the off-diagonal coupling elements have the same magnitudes as the diagonal elements – is not so straightforward. Under some circumstances it might be possible to perform an eigenanalysis to diagonalise parts of the system matrices, as was done in the case of the MDOF system in section 2.3.1. In general, however, strong coupling elements will remain in the off-diagonals. These result in a gradual evolution of the eigenvalues rather than rapid veering around the critical points. Furthermore, there is a distortion in the location of the points about which "veering" takes place. An example is shown in Figure 8, which concerns the same system considered in section 2.3.1, except that $k_n = 2k_0$ is no longer small.

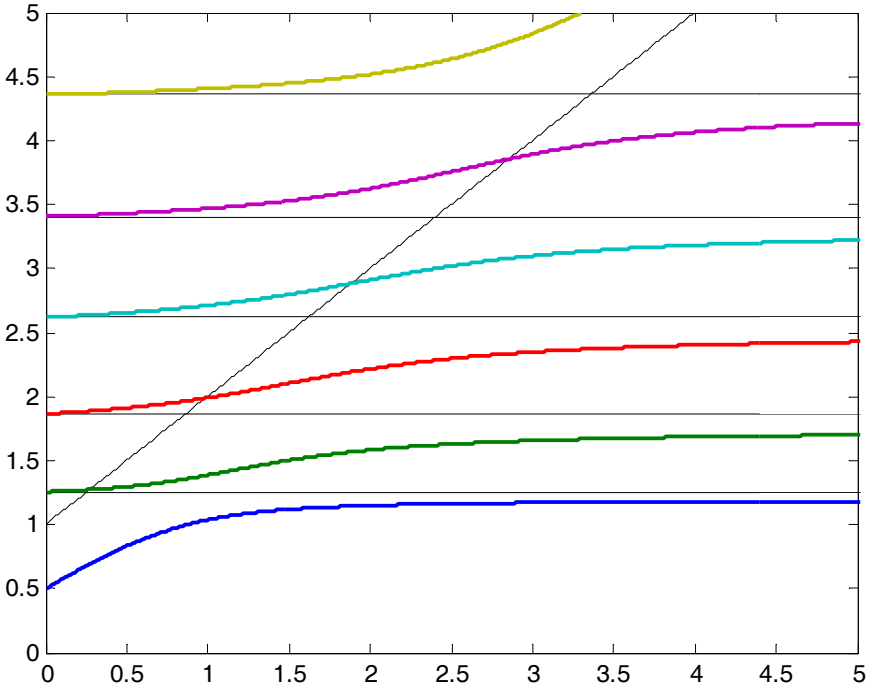


Figure 8: Strong coupling: veering in the 6-DOF system of section 2.3.1. but with $k_n = 2k_0$. Note a gradual evolution of the eigenvalues of the coupled system as opposed to veering around the critical points.

Finally, it could be that the off-diagonal terms depend on the random parameters so that there is variability in both diagonal and off-diagonal terms. This inevitably involves strong coupling, since the variable parameter(s) (e.g. stiffness) and the off-diagonal terms are of the same order of magnitude as the diagonal terms. There is consequently again an evolution of the eigenvalue spectrum rather than rapid veering, and a distortion of the points in the (p,λ) plane about which the evolution takes place. An example is shown in Figure 9, which concerns the plate example of section 3, except that the coupling spring is now the variable parameter and the spring connecting the mass to ground is removed. It can be noticed that for small values of ε weak coupling occurs, while as ε increases changes in the eigenvalues become less rapid.

5 Concluding remarks

This paper concerned mode veering in systems with a varying parameter. The phenomenon is commonplaces in systems with variable or random parameters. The emphasis was placed on weak coupling, in which case veering involves the rapid change in eigenvalues and eigenvectors of the coupled system around the critical points where the uncoupled-blocked eigenvalues intersect. For strong coupling or when the coupling elements themselves are variable there tends to be a less rapid change in the eigenproperties.

Acknowledgements

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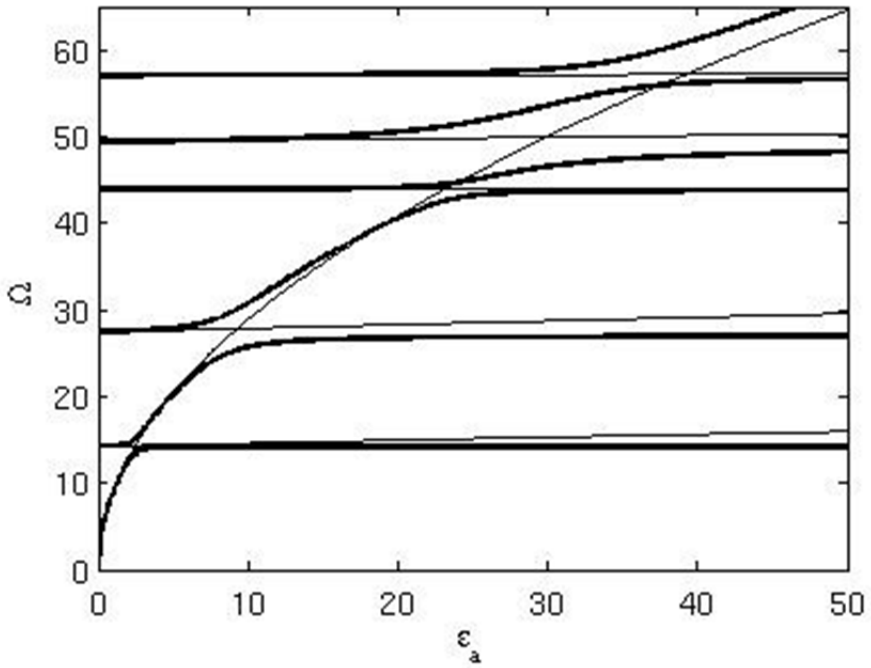


Figure 9: Strong coupling: veering in the plate system of section 3. Note that a gradual evolution of the eigenvalues of the coupled system as opposed to veering around the critical points occurs for increasing values of ε.

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